

Chapter 7

Image and Video Acquisition

IMP302m

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7.1 Image Scanning, Sampling, and Interpolation - Representations for the Sampled Image

- Image capture takes us from the continuous-parameter real world in which we live to the discrete-parameter, amplitude quantized domain of the digital devices that comprise an electronic imaging system.
- The process of converting from a continuous-parameter image to one that is discrete-parameter i.e., consists of an array of numbers, is referred to as *sampling*.
- The process of quantizing an image that is continuous in amplitude to one that takes on values from a finite set is called *quantization*.

7.1 Image Scanning, Sampling, and Interpolation - Representations for the Sampled Image

- Representations for the sampled image: sampling a continuous-space image $g_c(x, y)$ yields a discrete-space image

$$g_d(x, y) = g_c(mX, nY)$$

- Using 2D dirac delta function $\delta(x, y)$, sample value at fixed point

$$g_c(x_0, y_0) = \iint g_c(x, y) \delta(x - x_0, y - y_0) dx dy$$

or
$$g_c(x - x_0, y - y_0) = g_c(x, y) \star \delta(x - x_0, y - y_0)$$

- Get all samples of the image, the comb function is defined

$$\text{comb}_{X,Y}(x, y) = \sum_m \sum_n \delta(x - mX, y - nY)$$

- Continuous-parameter sampled image

$$g_s(x, y) = g_c(x, y) \text{comb}_{X,Y}(x, y) = \sum_m \sum_n g_d(m, n) \delta(x - mX, y - nY)$$

7.1 Image Scanning, Sampling, and Interpolation - Fourier analysis of image capture

- The forward and inverse versions of Continuous-space Fourier transform (CSFT) are given

$$G_c(u, v) = \iint g_c(x, y) e^{-j2\pi(ux+vy)} dx dy$$

$$g_c(x, y) = \iint G_c(u, v) e^{j2\pi(ux+vy)} du dv$$

where the units of frequency for (u, v) are cycles/unit distance.

7.1 Image Scanning, Sampling, and Interpolation - Fourier analysis of image capture

- The forward and inverse versions of Discrete-Space Fourier transform (DSFT) are given

$$G_d(u, v) = \sum_m \sum_n g_d(x, y) e^{-j2\pi(ux+vy)}$$

$$g_d(x, y) = \int_{-0.5}^{0.5} \int_{-0.5}^{0.5} G_d(u, v) e^{j2\pi(ux+vy)} du dv$$

where the units of frequency for (u, v) are cycles/pixel

7.1 Image Scanning, Sampling, and Interpolation - Fourier analysis of image capture

- For sampled signal

$$G_s(u, v) = G_c(u, v) \star \frac{1}{XY} \text{comb}_{\frac{1}{X}, \frac{1}{Y}}(u, v)$$

or

$$G_s(u, v) = \frac{1}{XY} \sum_k \sum_l G_c(u - \frac{k}{X}, v - \frac{l}{Y})$$

- Relates to Discrete-space Fourier Transform (DSFT)

$$G_s(u, v) = \frac{1}{XY} \sum_m \sum_n g_d(m, n) e^{-i2\pi(umX + vnY)}$$

- Therefore

$$G_s(u, v) = G_d(uX, vY)$$

7.1 Image Scanning, Sampling, and Interpolation

- What happens if the signal is sampled in limited area (NOT infinity)?

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

Downsampling and Decimation:

- To decrease the sampling rate of a digital image $f(m, n)$ by integer factors of $C \times D$: discard all but every C -th sample in the m direction and all but every D -th sample in the n direction

$$g_{\downarrow}(m, n) = f(Cm, Dn)$$

- In Fourier domain

$$\begin{aligned} G_{\downarrow}(u, v) &= \sum_m \sum_n f(Cm, Dn) e^{-i2\pi(um+vn)} \\ &= \sum_m \sum_n s_C(m) s_D(n) f(m, n) e^{-i2\pi(um/C+vn/D)} \end{aligned}$$

$s_C(m) = \begin{cases} 1, & m \% C = 0 \\ 0, & \text{otherwise} \end{cases}$

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

Downsampling and Decimation:

- Within one period, $s_C(m)$ consists of a single impulse

$$s_C(m) = \frac{1}{C} \sum_{k=0}^{C-1} e^{-i2\pi(mk/C)}$$

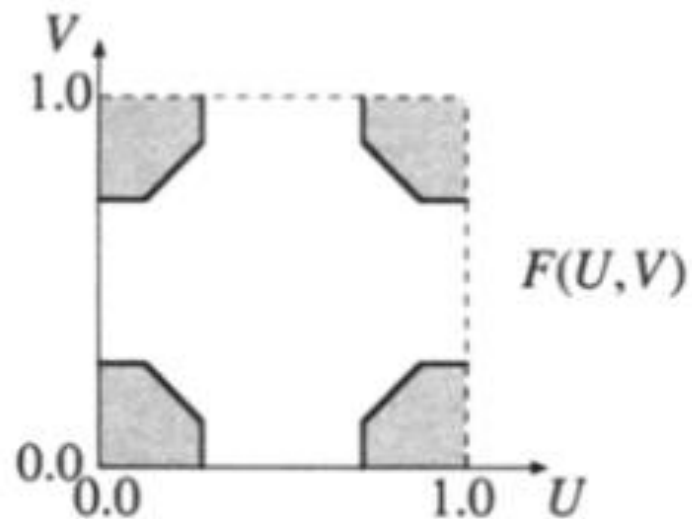
- Fourier transform of the downsampled signal is

$$\begin{aligned} G_{\downarrow}(u, v) &= \frac{1}{CD} \sum_{k=0}^{C-1} \sum_{l=0}^{D-1} \sum_m \sum_n f(m, n) e^{-i2\pi[(u-k)m/C + (v-l)n/D]} \\ &= \frac{1}{CD} \sum_k \sum_l F\left(\frac{u-k}{C}, \frac{v-l}{D}\right) \end{aligned}$$

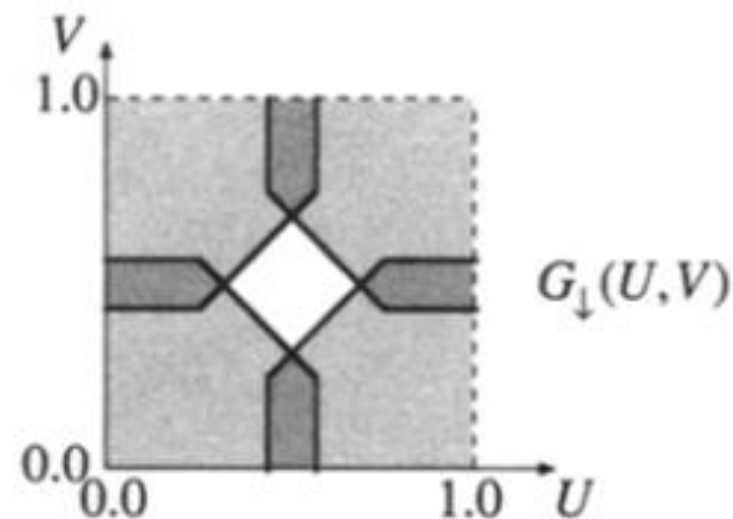
7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

Downsampling and Decimation

- downsampling the image causes the DSFT to be expanded by a factor of C in the u direction and a factor of D in the v direction.



(a)



(b)

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

Downsampling and Decimation

- The most important consequence of downsampling is the potential for additional aliasing, which will occur if $F(u, v) \neq 0$ for any $|u| \geq 1/(2C)$ or $|v| \geq 1/(2D)$.
- To prevent this, $f(m, n)$ is prefiltered prior to downsampling with a filter having frequency response

$$H(u, v) = CD \operatorname{rect}(CU, DV)$$

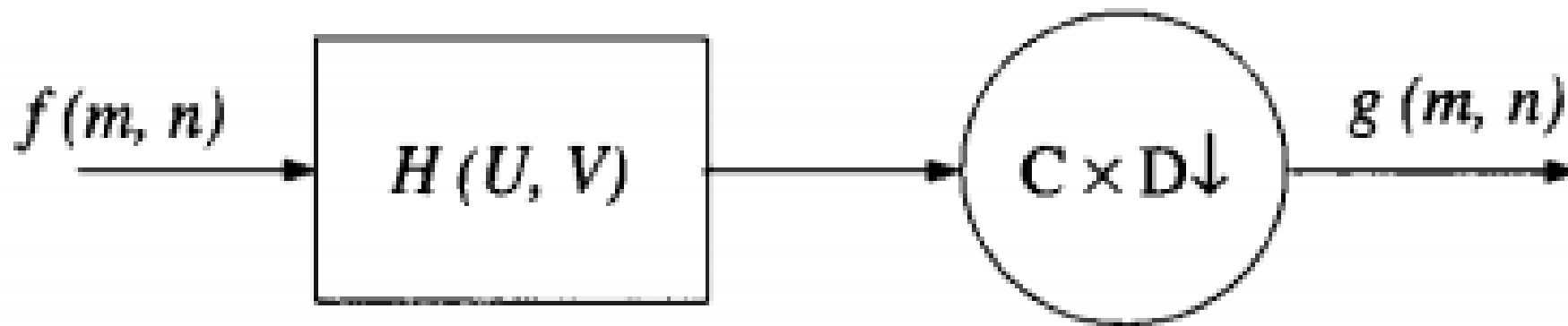
corresponding impulse response is

$$h(m, n) = \operatorname{sinc}(m/C, n/D)$$

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

Downsampling and Decimation

- The combination of a filter followed by a downsampler is called a *decimator*



$$g(m, n) = \sum_k \sum_l f(k, l) h(Cm - k, Dn - l)$$

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

- **Upsampling and Interpolation**
- Increase the sampling rate of an image $f(m, n)$ by integer factors C in the m direction and D in the n direction, it is helpful to start with an *upsampler* which inserts $C - 1$ zeros between each sample in the m direction and $D - 1$ zeros between each sample in the n direction

$$g_{\uparrow}(m, n) = \begin{cases} f(m, n), & m/C \text{ and } n/D \text{ are integers} \\ 0, & \text{otherwise} \end{cases}$$

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

- **Upsampling and Interpolation**

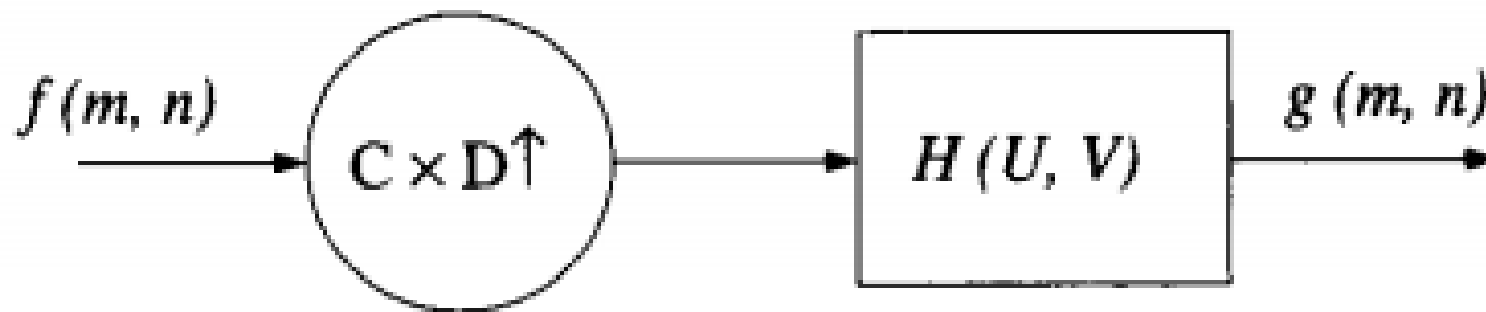
- In Fourier domain

$$\begin{aligned} G_{\downarrow}(u, v) &= \sum_m \sum_n s_C(m) s_D(n) f(m/C, n/D) e^{-i2\pi(um+vn)} \\ &= \sum_m \sum_n f(m, n) e^{-i2\pi(uCm+vDn)} \\ &= F(Cu, Dv) \end{aligned}$$

- Upsampling contracts the spectrum by a factor of C in the U direction and D in the V direction. There is no aliasing, because no information has been lost.

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

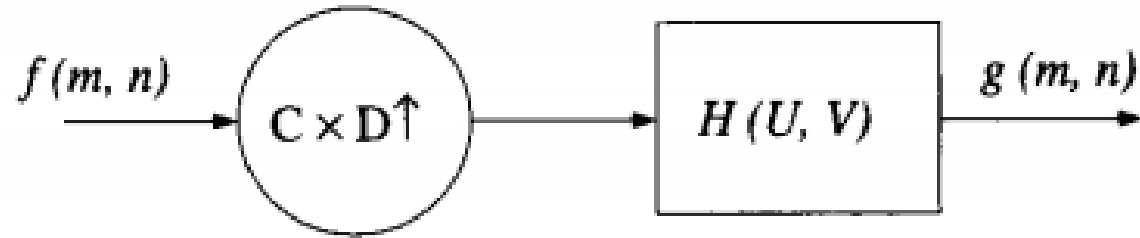
- **Upsampling and Interpolation**
- The combination of an upsampler followed by a filter is called an *interpolator*



$$g(m, n) = \sum_k \sum_l f(k, l) h(m - Ck, n - Dl)$$

7.1 Image Scanning, Sampling, and Interpolation - Sampling rate conversion

- Upsampling and Interpolation



- For the special case of the ideal lowpass filter

$$g(m, n) = \sum_k \sum_l f(k, l) \text{sinc}(m/C - k, n/D - l)$$

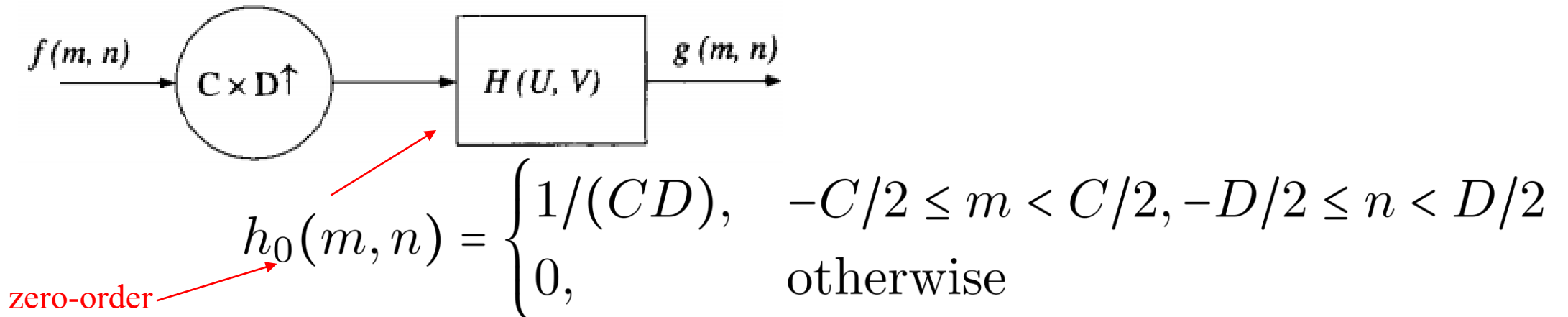
7.1 Image Scanning, Sampling, and Interpolation - Image Interpolation

- **Linear Filtering Approaches**

- The lowest level of computational complexity, *pixel replication* also known as *nearest neighbor interpolation* or *zero-order interpolation* is widely used in many applications

$$g(m, n) = f(\lfloor m/C \rfloor, \lfloor n/D \rfloor)$$

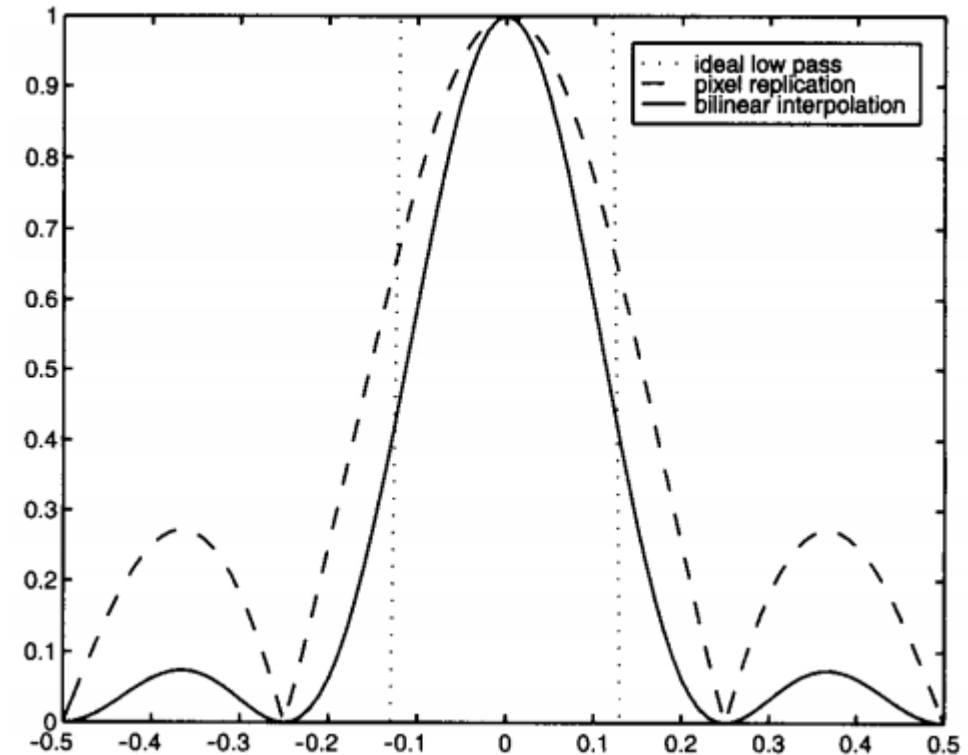
where $\lfloor \cdot \rfloor$ is rounding to the nearest integer



7.1 Image Scanning, Sampling, and Interpolation - Image Interpolation

- **Linear Filtering Approaches**
- Fourier transform of zero-order interpolation

$$H_0(u, v) = \frac{e^{i\pi(u+v)}}{CD} \left[\frac{\sin(\pi u C)}{\sin(\pi u)} \right] \left[\frac{\sin(\pi v D)}{\sin(\pi v)} \right]$$



7.1 Image Scanning, Sampling, and Interpolation - Image Interpolation

- **Linear Filtering Approaches**

- To obtain a smoother result, the additional samples between adjacent samples are linearly interpolate. The extension of this idea to 2D, called *bilinear interpolation*

$$g(m, n) = \alpha\beta f(\lfloor m/C \rfloor, \lfloor n/D \rfloor) + (1 - \alpha)\beta f(\lceil m/C \rceil, \lfloor n/D \rfloor) \\ + \alpha(1 - \beta)f(\lfloor m/C \rfloor, \lceil n/D \rceil) + (1 - \alpha)(1 - \beta)f(\lceil m/C \rceil, \lceil n/D \rceil)$$

where $\alpha = \lceil m/C \rceil - m/C$ and $\beta = \lceil n/D \rceil - n/D$

- First-order interpolation is

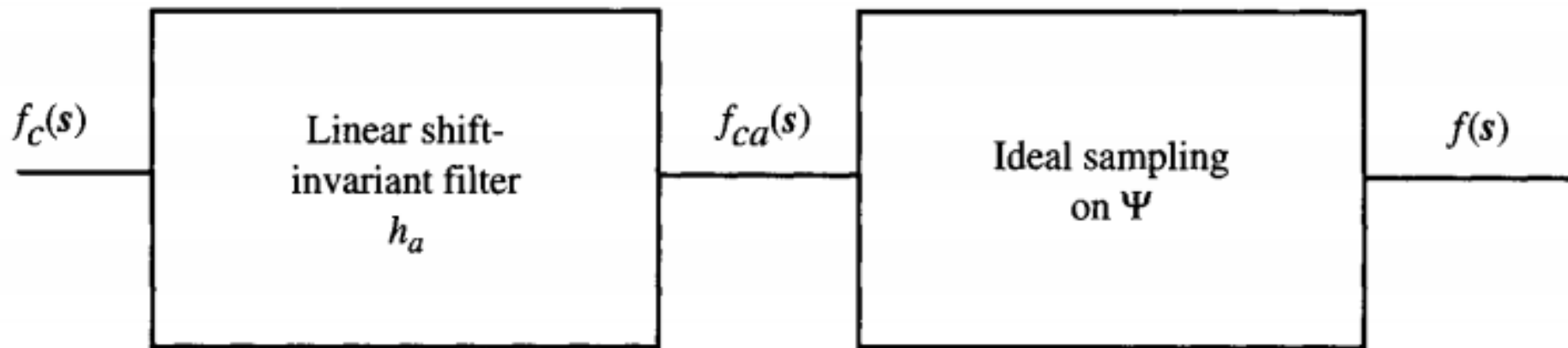
$$H_1(u, v) = |H_0(u, v)|^2 \overset{\mathfrak{F}}{\leftarrow} h_1(m, n) = \begin{cases} (1 - |m/C|)(1 - |n/D|), & |m| < C \text{ and } |n| < D \\ 0, & \text{otherwise} \end{cases} \\ = h_0(m, n) \star h_0(-m, -n)$$

7.2 Video Sampling and Interpolation

- The initial acquisition and scanning is particularly critical because it determines what information is contained in the original data.
- The acquisition process can be modeled as a continuous space-time prefiltering followed by ideal sampling on a given sampling structure.
- The sampling structure determines the amount of spatio-temporal information that the sampled signal can carry, while the prefiltering serves to limit the amount of aliasing
- The desired display characteristics are closely related to the properties of the human visual system

7.2 Video Sampling and Interpolation

- System for sampling a time-varying image



7.2 Video Sampling and Interpolation - Spatiotemporal Sampling Structures

- Definitions:
 - A continuous time-varying image $f_c(x, y, t)$ is a function of two spatial dimensions x and y and time t , usually observed in a rectangular spatial window $\mathcal{W}$ over some time interval T .
 - The spatiotemporal region $\mathcal{W} \times T$ is denoted $\mathcal{W}_T$
 - The spatial window is of dimension $pw \times ph$ where pw is the picture width and ph is the picture height
- The image f_c can be sampled in one, two or three dimensions.
- Always sampled in at least the temporal dimension, producing an *image sequence*

7.2 Video Sampling and Interpolation - Spatiotemporal Sampling Structures

- A lattice Λ in D dimensions is a discrete set of points that can be expressed as the set of all linear combinations with integer coefficients of D linearly independent vectors where $D = \{1, 2, 3\}$
- For one-dimensional temporal sampling, the 1D lattice is $\Lambda_t = \{nT | n \in \mathbb{Z}\}$ where T is the frame period. The sampling structure is then

$$\mathcal{W} \times \Lambda_t = \{(\mathbf{x}, t) | \mathbf{x} \in \mathcal{W}, t \in \Lambda_t\}$$

- For two-dimensional vertical-temporal sampling (scanning) using a 2D lattice Λ_{yt} . The sampling structure is $\mathcal{W}_T \cap (\mathcal{H} \times \Lambda_{yt})$ where $\mathcal{H}$ is a one-dimensional subspace of $\mathbb{R}^3$ parallel to the scanning lines

7.2 Video Sampling and Interpolation - Spatiotemporal Sampling Structures

- The effect of sampling time-varying images on a lattice
- For a continuous signal f_c the Fourier transform is given by

$$F_c(u, v, w) = \iiint f_c(x, y, t) e^{-i2\pi(ux+vy+wt)} dx dy dt$$

or

$$F_c(\mathbf{u}) = \int_{\mathcal{W}_T} f_c(\mathbf{s}) e^{-i2\pi \mathbf{u} \cdot \mathbf{s}} d\mathbf{s}$$

- A discrete signal has a lattice Fourier transform

$$F(\mathbf{u}) = \sum_{\mathbf{s} \in \Lambda} f(\mathbf{s}) e^{-i2\pi \mathbf{u} \cdot \mathbf{s}}$$

7.2 Video Sampling and Interpolation - Sampling and Reconstruction

- The light arriving on the sensor is collected and weighted in space and time by the sensor aperture $a(\mathbf{s})$, space and time invariant, to give the output

$$f_{ca}(\mathbf{s}) = \int_{\mathbb{R}^3} f_c(\mathbf{s} + \mathbf{s}') a(\mathbf{s}') d\mathbf{s}'$$

- The resulting signal $f_{ca}(\mathbf{s})$ is then sampled in an ideal fashion on the sampling structure Ψ

$$f(\mathbf{s}) = f_{ca}(\mathbf{s}), \quad \mathbf{s} \in \Psi$$

- The aperture weighting is a linear shift-invariant filtering operation

$$f_{ca}(\mathbf{s}) = \int_{\mathbb{R}^3} f_c(\mathbf{s} - \mathbf{s}') h_a(\mathbf{s}') d\mathbf{s}', \quad h_a(\mathbf{s}) = a(-\mathbf{s})$$

7.2 Video Sampling and Interpolation - Sampling and Reconstruction

- In practical systems, the reconstruction is achieved by

$$\hat{f}_{ca}(\mathbf{s}) = \sum_{\mathbf{s}' \in \Lambda} f(\mathbf{s}') d(\mathbf{s} - \mathbf{s}')$$

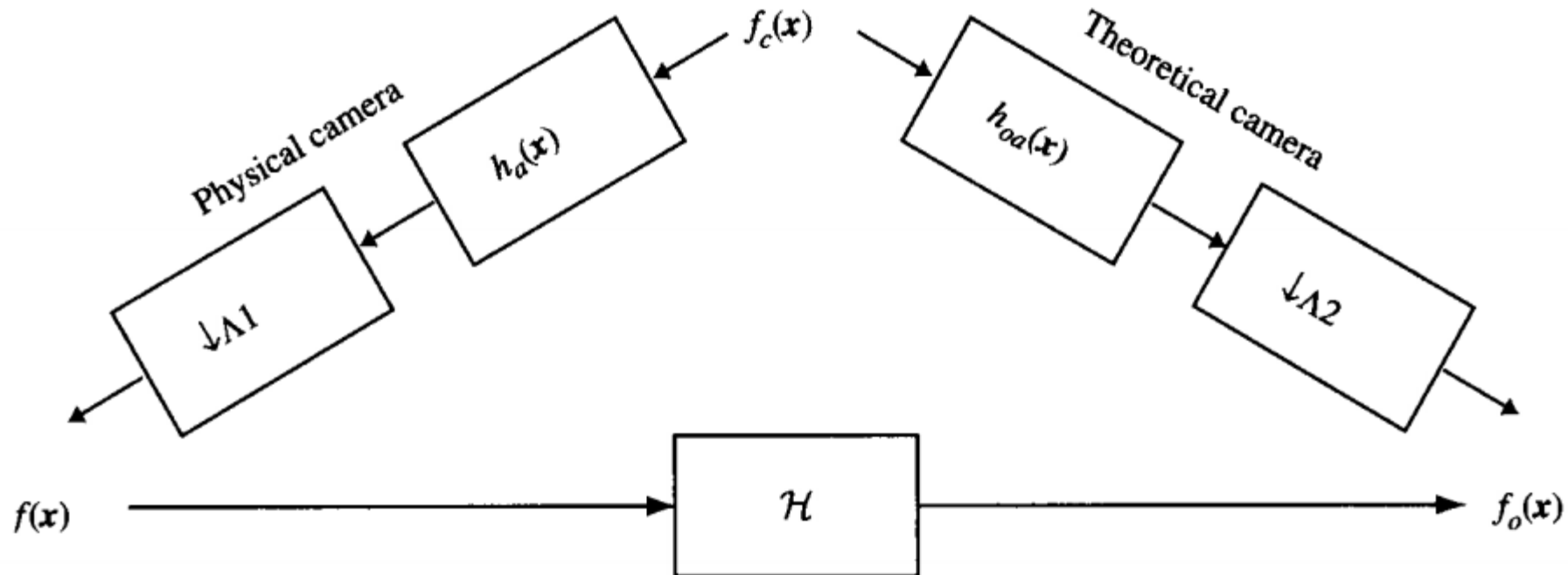
where $d(\mathbf{s})$ is the display aperture

- If sampling is performed in only one or two dimensions, the spectrum is replicated in the corresponding frequency dimensions.
 - Temporal sampling

$$F(\mathbf{u}, w) = \frac{1}{T} \sum_{l=-\infty}^{\infty} F_{ca} \left(\mathbf{u}, w + \frac{l}{T} \right)$$

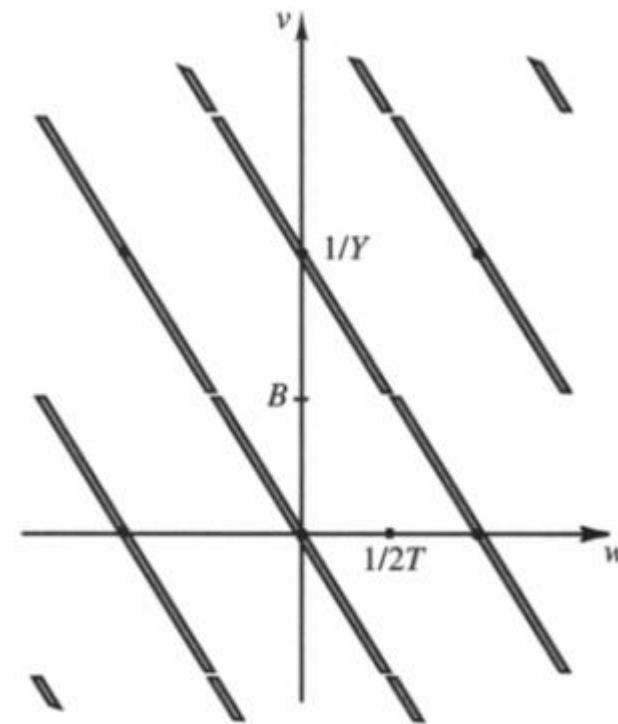
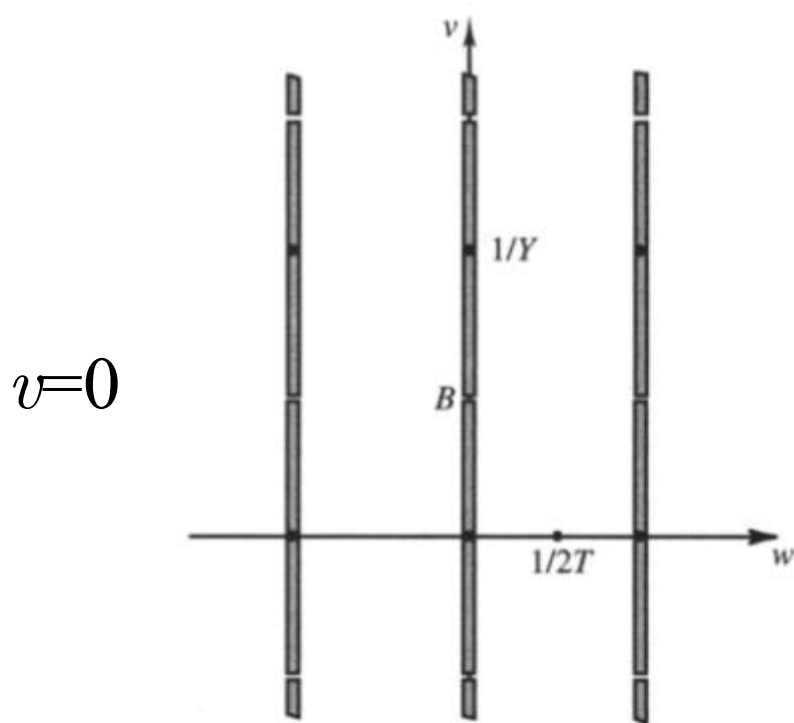
7.2 Video Sampling and Interpolation – Frame-Rate Conversion

- Acquisition models for the observed signal $f(x)$ on Λ_1 and the desired output signal $f_0(x)$ on Λ_2



7.2 Video Sampling and Interpolation – Frame-Rate Conversion

- Vertical-temporal projection of spectrum of vertical-temporal sampled time-varying image with progressive scanning



$$v=1/2TB$$

7.2 Video Sampling and Interpolation – Frame-Rate Conversion

- **Pure temporal interpolation**
- A typical application for this is increasing the frame rate in motion picture film from 24 frames/s to 48 or 60 frames/s, giving significantly better motion rendition. Using linear filtering, the interpolated image sequence is given by

$$f_0(\mathbf{x}, nT_2) = \sum_m f(\mathbf{x}, mT_1) h(nT_2 - mT_1)$$

- If the temporal spectrum of the underlying continuous time-varying image satisfies the Nyquist criterion, the output points can be computed by ideal *sinc* interpolation:

$$h(t) = \frac{\sin(\pi t/T_1)}{\pi t/T_1} = \text{sinc}(\pi t/T_1)$$

7.2 Video Sampling and Interpolation – Frame-Rate Conversion

- **Motion-compensated interpolation**
- Discussion: Why do we need motion-compensated interpolation?

Summary

- Image Scanning, Sampling, and Interpolation
 - ✓ Representations for the Sampled Image
 - ✓ Fourier analysis of image capture
 - ✓ Sampling rate conversion
- Video Sampling and Interpolation